

Below is **Part 1: Physical & Chemical Changes -True or False** questions.

topic 1 – TRUE OR FALSE

Physical & Chemical Changes

1. Rotting tomatoes is a chemical change because new substances are formed.

Answer: True

Explanation: Rotting involves decomposition, producing new substances.

2. Crushing chunks of ice is a chemical change because its shape changes.

Answer: False

Explanation: Crushing ice only changes its size and shape. The substance remains water.

3. A chemical change produces one or more new substances.

Answer: True

Explanation: The chemical composition changes during a chemical change.

4. Photosynthesis is a physical change because plants only absorb sunlight.

Answer: False

Explanation: Photosynthesis is a chemical reaction that produces glucose and oxygen.

5. Gas bubbles produced when vinegar reacts with baking soda are evidence of a chemical reaction.

Answer: True

Explanation: The bubbles are carbon dioxide gas formed during the reaction.

6. Curdling of milk is considered evidence of gas formation.

Answer: False

Explanation: It is evidence of precipitate formation, where solid curds are produced.

7. If a container becomes hot after quicklime is mixed with water, a chemical reaction has occurred.

Answer: True

Explanation: The release of heat indicates an exothermic chemical reaction.

8. Simply cutting paper is evidence that a chemical reaction occurred.

Answer: False

Explanation: Cutting only changes the paper's shape, not its composition.

9. When two clear liquids produce a solid that settles at the bottom, a precipitate has formed.

Answer: True

Explanation: A precipitate is an insoluble solid formed during a reaction.

10. Reactants are written on the right side of a chemical equation.

Answer: False

Explanation: Reactants are written on the left side of the reaction arrow.

11. Products are the new substances formed during a chemical reaction.

Answer: True

Explanation: Products appear on the right side of the chemical equation.

12. The symbol **(aq)** means the substance is in the gaseous state.

Answer: False

Explanation: (aq) means the substance is dissolved in water.

13. The symbol **Δ** above a reaction arrow means heat is supplied to the reaction.

Answer: True

Explanation: Delta (Δ) indicates that heating is required.

14. A downward arrow (↓) beside a product indicates that a gas is released.

Answer: False

Explanation: A downward arrow indicates the formation of a precipitate.

15. The plus (+) sign in a chemical equation means the substances are mixed or react together.

Answer: True

Explanation: It is read as "reacts with" or "and."

16. A combination reaction follows the general equation **A + B → AB**.

Answer: True

Explanation: Two or more reactants combine to form a single product.

17. The reaction **2Na + Cl₂ → 2NaCl** is an example of decomposition.

Answer: False

Explanation: It is a combination (synthesis) reaction because two reactants form one product.

18. The reaction **CaCO₃ → CaO + CO₂** is a decomposition reaction.

Answer: True

Explanation: One compound breaks down into two simpler substances.

19. The general equation **AB → A + B** represents a combination reaction.

Answer: False

Explanation: It represents decomposition.

20. In a single displacement reaction, one element replaces another element in a compound.

Answer: True

Explanation: This is the defining characteristic of single displacement.

21. Double displacement reactions involve one element replacing another element.

Answer: False

Explanation: They involve two compounds exchanging ions.

22. The reaction **AgNO₃ + NaCl → AgCl + NaNO₃** is a double displacement reaction.

Answer: True

Explanation: The ions exchange partners to form new compounds.

23. Oxygen gas is always a reactant in combustion reactions.

Answer: True

Explanation: Combustion requires oxygen to occur.

24. Combustion of hydrocarbons always produces salt and water.

Answer: False

Explanation: It produces carbon dioxide and water.

25. Neutralization reactions produce salt and water.

Answer: True

Explanation: They occur when an acid reacts with a base.

26. Neutralization is classified as a special type of combination reaction.

Answer: False

Explanation: It is a special type of double displacement reaction.

27. Melting wax is an example of a physical change.

Answer: True

Explanation: Only the state changes; no new substance is formed.

28. Salt dissolving in water is evidence of a chemical reaction.

Answer: False

Explanation: Dissolving salt is a physical change.

29. In the equation $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$, magnesium oxide is the product.

Answer: True

Explanation: Products are found on the right side of the arrow.

30. Fuel reacting with oxygen to produce carbon dioxide and water is called combustion.

Answer: True

Explanation: This is the general form of a combustion reaction.

31. The arrow ($\rightarrow$) in a chemical equation means "yields" or "produces."

Answer: True

Explanation: It separates reactants from products.

32. The reaction $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ is a synthesis reaction.

Answer: False

Explanation: It is a decomposition reaction because one compound breaks apart.

33. Double displacement reactions involve the exchange of ions between two compounds.

Answer: True

Explanation: The ions switch partners to form new compounds.

34. Iron reacting with sulfur to form iron sulfide is an example of synthesis.

Answer: True

Explanation: Two reactants combine to form one compound.

35. The symbol (g) or an upward arrow ($\uparrow$) indicates that a solid has formed.

Answer: False

Explanation: It indicates that a gas is produced.

TRUE OR FALSE : topic 3 :Acids, Bases, and Salts

1. All acids have a sour taste and turn blue litmus paper red.

Answer: True

Explanation: These are characteristic properties of acids.

2. Bases turn blue litmus paper red.

Answer: False

Explanation: Bases turn **red litmus paper blue**.

3. Pure water has a pH of 7, making it neutral.

Answer: True

Explanation: A pH of 7 indicates a neutral solution.

4. Phenolphthalein is pink in acidic solutions.

Answer: False

Explanation: Phenolphthalein is colorless in acids and pink in bases.

5. Acids release hydrogen ions (H⁺) when dissolved in water.

Answer: True

Explanation: This follows the Arrhenius definition of acids.

6. Bases release hydrogen ions (H⁺) in water.

Answer: False

Explanation: Bases release hydroxide ions (OH⁻).

7. Sodium chloride (NaCl) is the chemical name for common table salt.

Answer: True

Explanation: Sodium chloride is a neutral salt commonly used as table salt.

8. Hydrochloric acid found in the stomach has the formula H₂SO₄.

Answer: False

Explanation: The correct formula is HCl.

9. Sodium hydroxide (NaOH) is a strong base commonly known as lye.

Answer: True

Explanation: NaOH is also called caustic soda or lye.

10. Methyl orange is yellow in acids and red in bases.

Answer: False

Explanation: Methyl orange is **red in acids** and **yellow in bases**.

11. Turmeric is considered a natural acid-base indicator.

Answer: True

Explanation: Turmeric changes color in basic solutions.

12. A solution with a pH of 7 is acidic.

Answer: False

Explanation: A pH of 7 is neutral.

13. Vinegar contains acetic acid.

Answer: True

Explanation: Acetic acid gives vinegar its sour taste.

14. Milk of magnesia contains sodium hydroxide.

Answer: False

Explanation: It contains magnesium hydroxide, $\text{Mg}(\text{OH})_2$.

15. Blue litmus paper remains blue when placed in a basic solution.

Answer: True

Explanation: Bases do not change the color of blue litmus paper.

16. Hydrochloric acid, vinegar, and calamansi juice are all examples of acids.

Answer: True

Explanation: All three exhibit acidic properties.

17. Sodium hydroxide, bleach, and milk of magnesia are examples of bases.

Answer: True

Explanation: These substances are classified as bases.

18. Antacids relieve hyperacidity because they are weak acids.

Answer: False

Explanation: Antacids are mild bases that neutralize excess stomach acid.

19. A solution with a pH of 4 is acidic.

Answer: True

Explanation: Any pH below 7 is acidic.

20. A solution with a pH of 11 is acidic.

Answer: False

Explanation: A pH above 7 indicates a basic or alkaline solution.

21. Among pH 1, pH 3, and pH 6, pH 6 is the weakest acid.

Answer: True

Explanation: It is closest to neutral (pH 7).

22. A neutralization reaction produces salt and water from an acid and a base.

Answer: True

Explanation: This is the general equation for neutralization.

23. Turmeric turns reddish-brown when exposed to a base such as soap.

Answer: True

Explanation: This color change indicates the presence of a base.

24. Onion and vanilla are examples of synthetic indicators.

Answer: False

Explanation: They are olfactory indicators because they change smell in acids and bases.

25. Tooth enamel begins to dissolve when the pH in the mouth drops below 5.5.

Answer: True

Explanation: Low pH increases the risk of tooth decay.

26. Sodium carbonate (Na_2CO_3) forms an acidic solution in water.

Answer: False

Explanation: It forms a basic solution.

27. Ammonium chloride (NH_4Cl) forms an acidic solution when dissolved in water.

Answer: True

Explanation: NH_4Cl is an acidic salt.

28. Dry hydrogen chloride gas changes dry blue litmus paper to red immediately.

Answer: False

Explanation: Water is needed for HCl to release hydrogen ions.

29. Each pH unit represents a tenfold change in hydrogen ion concentration.

Answer: True

Explanation: The pH scale is logarithmic.

30. In the order pH 2, pH 6, pH 7, pH 9, and pH 12, pH 12 is the strongest acid.

Answer: False

Explanation: pH 12 is the strongest base, while pH 2 is the strongest acid.

31. Sodium chloride (NaCl) is produced from a strong acid and a strong base and is generally neutral.

Answer: True

Explanation: HCl reacts with NaOH to produce NaCl and water.

32. Acidic and basic solutions can conduct electricity because they contain mobile ions.

Answer: True

Explanation: The movement of ions allows electric current to flow.

33. When diluting concentrated acid, water should always be poured into the acid.

Answer: False

Explanation: Acid should always be added slowly to water to prevent splashing.

34. Acetic acid (CH_3COOH) is the weak acid present in vinegar.

Answer: True

Explanation: It is responsible for vinegar's acidic properties.

35. Baking soda reacts with acids to release oxygen gas.

Answer: False

Explanation: It releases carbon dioxide (CO_2) gas during the reaction.

Topic 4 – TRUE OR FALSE : Chemical Equations

1. Reactants are written on the left side of a chemical equation.

Answer: True

Explanation: Reactants are the starting substances before the reaction occurs.

2. Products are found on the left side of the reaction arrow.

Answer: False

Explanation: Products are written on the right side of the arrow because they are formed after the reaction.

3. The symbol **(aq)** means a substance is dissolved in water.

Answer: True

Explanation: "Aqueous" means the substance is in a water solution.

4. The symbol **(s)** represents a gas.

Answer: False

Explanation: (s) stands for solid.

5. The symbol **(g)** indicates that a substance is in the gaseous state.

Answer: True

Explanation: (g) is the standard symbol for gases.

6. The Law of Conservation of Mass states that mass can be created during a chemical reaction.

Answer: False

Explanation: Mass is neither created nor destroyed; the total mass of reactants equals the total mass of products.

7. The arrow ($\rightarrow$) in a chemical equation is read as "yields" or "produces."

Answer: True

Explanation: It separates the reactants from the products.

8. The plus (+) sign between reactants means "reacts with" or "and."

Answer: True

Explanation: It indicates that substances are combined in the reaction.

9. A balanced chemical equation has the same number of atoms of each element on both sides.

Answer: True

Explanation: This follows the Law of Conservation of Mass.

10. The balanced equation for forming water is $\text{H}_2 + \text{O} \rightarrow \text{H}_2\text{O}$.

Answer: False

Explanation: The correct balanced equation is $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$.

11. The symbol **(l)** indicates that a substance is in the liquid state.

Answer: True

Explanation: (l) stands for liquid.

12. A coefficient tells the number of molecules or moles involved in a reaction.

Answer: True

Explanation: It multiplies the entire chemical formula.

13. A subscript tells the number of molecules present in a reaction.

Answer: False

Explanation: A subscript tells the number of atoms of a particular element in one molecule or formula unit.

14. Carbon dioxide has the chemical formula CO₂.

Answer: True

Explanation: It contains one carbon atom and two oxygen atoms.

15. In CaCl₂, the subscript "2" refers to the number of chlorine atoms.

Answer: True

Explanation: There are two chlorine atoms bonded to one calcium atom.

16. When balancing chemical equations, only coefficients should be changed.

Answer: True

Explanation: Changing subscripts changes the identity of the substance.

17. In 3H₂O, the number 3 is a subscript.

Answer: False

Explanation: The number 3 is a coefficient, while the 2 is a subscript.

18. The equation $\text{Ca} + \text{Cl}_2 \rightarrow \text{CaCl}_2$ is already balanced.

Answer: True

Explanation: There is one calcium atom and two chlorine atoms on both sides.

19. In 2H₂SO₄, there are four hydrogen atoms in total.

Answer: True

Explanation: Two molecules each contain two hydrogen atoms ($2 \times 2 = 4$).

20. In 4Al₂O₃, there are eight oxygen atoms altogether.

Answer: False

Explanation: There are 12 oxygen atoms ($4 \times 3 = 12$).

21. In the equation $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$, the product side contains two chlorine atoms.

Answer: True

Explanation: The subscript 2 in MgCl₂ represents two chlorine atoms.

22. The first step in balancing a chemical equation is to count the atoms on both sides.

Answer: True

Explanation: Counting atoms helps determine which coefficients need adjustment.

23. Chemical equations are balanced mainly to make them easier to read.

Answer: False

Explanation: They are balanced to obey the Law of Conservation of Mass.

24. If gas escapes from an open container during a reaction, the Law of Conservation of Mass is still valid.

Answer: True

Explanation: The gas escaped the container, but the total mass of all matter remains unchanged.

25. In the equation $\text{N}_2 + \text{H}_2 \rightarrow \text{NH}_3$, both nitrogen and hydrogen atoms are initially unbalanced.

Answer: True

Explanation: Both elements must be balanced by adjusting coefficients.

26. In $5\text{C}_6\text{H}_{12}\text{O}_6$, there are 30 carbon atoms in total.

Answer: True

Explanation: Five molecules each contain six carbon atoms ($5 \times 6 = 30$).

27. Changing subscripts is an acceptable method for balancing chemical equations.

Answer: False

Explanation: Changing subscripts changes the chemical substance itself.

28. The balanced equation for forming water is $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$.

Answer: True

Explanation: This equation has equal numbers of hydrogen and oxygen atoms on both sides.

29. The balanced equation $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ correctly follows the Law of Conservation of Mass.

Answer: True

Explanation: Both sides contain two nitrogen atoms and six hydrogen atoms.

30. The balanced equation $2\text{Al} + \text{O}_2 \rightarrow \text{Al}_2\text{O}_3$ correctly balances aluminum and oxygen atoms.

Answer: False

Explanation: The correct balanced equation is $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$.

31. The balanced equation $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$ obeys the Law of Conservation of Mass.

Answer: True

Explanation: The number of each type of atom is equal on both sides.

32. In the balanced equation $2\text{C}_2\text{H}_6 + 7\text{O}_2 \rightarrow 4\text{CO}_2 + 6\text{H}_2\text{O}$, there are 14 oxygen atoms on the reactant side.

Answer: True

Explanation: Seven oxygen molecules contain 14 oxygen atoms.

33. If 12 g of carbon reacts completely with 32 g of oxygen, the total mass of carbon dioxide produced is 44 g.

Answer: True

Explanation: The masses of the reactants add up to the mass of the products.

34. Balanced chemical equations show that atoms are rearranged, not created or destroyed.

Answer: True

Explanation: This is the principle behind the Law of Conservation of Mass.

35. Changing subscripts while balancing an equation is acceptable because it does not change the identity of a substance.

Answer: False

Explanation: Changing subscripts changes the chemical formula and creates a different substance.

TOPIC 5 – TRUE OR FALSE

Chemical Equations

1. Reactants are the substances present before a chemical reaction begins.

Answer: True

Explanation: Reactants are the starting materials found on the left side of the equation.

2. Products are written on the left side of the reaction arrow.

Answer: False

Explanation: Products are written on the right side because they are formed after the reaction.

3. The reaction arrow (→) means "reacts to form" or "produces."

Answer: True

Explanation: It shows the direction of the chemical reaction.

4. The symbol (s) indicates that a substance is dissolved in water.

Answer: False

Explanation: (s) means the substance is in the solid state.

5. The symbol (aq) means the substance is dissolved in water.

Answer: True

Explanation: "Aqueous" refers to a solution in water.

6. A coefficient is the small number written below an element symbol.

Answer: False

Explanation: A coefficient is the large number placed before a chemical formula.

7. A subscript tells the number of atoms of an element in one molecule or formula unit.

Answer: True

Explanation: Subscripts are part of a compound's chemical formula.

8. The Law of Conservation of Mass states that the total mass of the reactants equals the total mass of the products.

Answer: True

Explanation: Atoms are rearranged but are neither created nor destroyed.

9. A combination reaction produces one product from two or more reactants.

Answer: True

Explanation: Its general form is $A + B \rightarrow AB$.

10. A decomposition reaction combines two substances to form one product.

Answer: False

Explanation: Decomposition involves one compound breaking into simpler substances.

11. A catalyst speeds up a reaction without being used up or permanently changed.

Answer: True

Explanation: Catalysts lower the activation energy of a reaction.

12. An exothermic reaction absorbs heat from its surroundings.

Answer: False

Explanation: Exothermic reactions release heat or light energy.

13. Endothermic reactions absorb energy from their surroundings.

Answer: True

Explanation: These reactions require an input of energy.

14. In a single displacement reaction, two compounds exchange ions.

Answer: False

Explanation: Single displacement involves one element replacing another in a compound.

15. Double displacement reactions involve two compounds exchanging ions.

Answer: True

Explanation: The products are formed by swapping ions between compounds.

16. Oxygen gas is always a reactant in combustion reactions.

Answer: True

Explanation: Combustion requires oxygen to occur.

17. Chemical equations are used only to represent physical changes.

Answer: False

Explanation: They represent chemical changes where new substances are formed.

18. The symbol Δ above a reaction arrow means that heat is applied.

Answer: True

Explanation: Delta (Δ) indicates heating during the reaction.

19. Rusting of iron is a physical change because no new substance is produced.

Answer: False

Explanation: Rusting is a chemical change that forms iron oxide.

20. Word equations use names of substances, while chemical equations use formulas and symbols.

Answer: True

Explanation: Chemical equations provide a more precise representation of reactions.

21. An insoluble solid formed during a reaction is called a precipitate.

Answer: True

Explanation: Precipitates are often indicated by (s) or a downward arrow ($\downarrow$).

22. Subscripts can be changed when balancing equations because they do not affect the identity of a substance.

Answer: False

Explanation: Changing a subscript changes the chemical formula and creates a different substance.

23. A coefficient applies to the entire chemical formula that follows it.

Answer: True

Explanation: It multiplies every atom in the formula.

24. A typical chemical equation directly shows how fast a reaction occurs.

Answer: False

Explanation: It shows the substances involved, not the reaction rate.

25. A reversible reaction is represented by the symbol $\rightleftharpoons$.

Answer: True

Explanation: This symbol indicates the reaction can proceed in both directions.

26. Even if a chemical equation is not balanced, it can still identify the reactants, products, and reaction type.

Answer: True

Explanation: Balancing affects quantity, not the identity of the substances.

27. The reaction arrow ($\rightarrow$) means the reactants and products are always equal in every way.

Answer: False

Explanation: The arrow indicates the direction of the reaction, not equality.

28. The symbol (g) beside a product indicates that a gas is produced.

Answer: True

Explanation: (g) represents the gaseous state.

29. Chemical formulas are used because they show the types and numbers of atoms in a substance.

Answer: True

Explanation: Formulas provide exact information about chemical composition.

30. The main purpose of a chemical equation is to summarize a chemical reaction using standard symbols and formulas.

Answer: True

Explanation: It clearly shows the reactants, products, and direction of the reaction.

Certainly! Here's **Part 6** in the **same format** as Parts 1–5.

Topic 6 – TRUE OR FALSE

Rates of Chemical Reactions

1. Reaction rate refers to how fast a chemical reaction takes place.

Answer: True

Explanation: The rate of reaction measures how quickly reactants are converted into products.

2. Increasing the temperature usually slows down the rate of a chemical reaction.

Answer: False

Explanation: Higher temperatures increase particle movement, leading to more frequent and energetic collisions.

3. A catalyst speeds up a chemical reaction without being permanently consumed.

Answer: True

Explanation: Catalysts lower the activation energy needed for a reaction.

4. Endothermic reactions release heat to their surroundings.

Answer: False

Explanation: Endothermic reactions absorb heat from their surroundings.

5. Exothermic reactions release heat energy into the surroundings.

Answer: True

Explanation: Combustion and neutralization are common examples of exothermic reactions.

6. Increasing the surface area of a solid reactant generally increases the reaction rate.

Answer: True

Explanation: More surface area allows more particles to collide at the same time.

7. The fire triangle is made up of fuel, oxygen, and carbon dioxide.

Answer: False

Explanation: The fire triangle consists of fuel, oxygen, and heat.

8. Refrigeration preserves food by slowing down chemical reactions and microbial activity.

Answer: True

Explanation: Lower temperatures reduce the rate of spoilage reactions.

9. An inhibitor increases the rate of a chemical reaction.

Answer: False

Explanation: Inhibitors slow down or prevent chemical reactions.

10. Burning wood is an example of an exothermic reaction.

Answer: True

Explanation: Burning releases heat and light energy.

11. Activation energy is the minimum amount of energy required for a reaction to occur.

Answer: True

Explanation: Reacting particles must overcome this energy barrier.

12. According to collision theory, every collision between particles produces a reaction.

Answer: False

Explanation: Only effective collisions with enough energy and proper orientation produce reactions.

13. Crushing a tablet into powder usually makes it dissolve faster.

Answer: True

Explanation: Crushing increases surface area, allowing faster interaction with the solvent.

14. Larger pieces of charcoal burn faster than powdered charcoal.

Answer: False

Explanation: Powdered charcoal has greater surface area and burns more quickly.

15. Lowering the temperature decreases the reaction rate of most chemical reactions.

Answer: True

Explanation: Lower temperatures reduce particle movement and collision frequency.

16. Powdered calcium carbonate reacts more quickly with hydrochloric acid than marble chips.

Answer: True

Explanation: Powder has a much greater surface area exposed to the acid.

17. Increasing the concentration of reactants generally increases the rate of reaction.

Answer: True

Explanation: More particles are available to collide each second.

18. Enzymes are biological catalysts found in living organisms.

Answer: True

Explanation: They speed up biochemical reactions without being consumed.

19. Water extinguishes most fires mainly by removing oxygen from the flame.

Answer: False

Explanation: Water primarily cools the fuel below its ignition temperature.

20. A fire blanket extinguishes fire by cutting off the oxygen supply.

Answer: True

Explanation: Smothering the fire prevents oxygen from reaching it.

21. Painting iron helps prevent rusting by reducing contact with oxygen and water.

Answer: True

Explanation: The paint acts as a protective barrier.

22. Salting and sugaring preserve food by reducing the amount of available water for microorganisms.

Answer: True

Explanation: Microorganisms need water to survive and multiply.

23. Canning preserves food mainly by increasing its surface area.

Answer: False

Explanation: Canning uses heat to destroy microorganisms and seals the food to prevent contamination.

24. Stirring reactants can increase the rate of a chemical reaction.

Answer: True

Explanation: Stirring brings reactant particles into contact more frequently.

25. Photosynthesis is an endothermic reaction because it absorbs energy from sunlight.

Answer: True

Explanation: Light energy is required to convert carbon dioxide and water into glucose.

26. Higher temperatures increase reaction rates only because particles collide more often.

Answer: False

Explanation: They also increase the number of particles with enough energy to overcome activation energy.

27. Catalysts provide an alternative reaction pathway with lower activation energy.

Answer: True

Explanation: This allows reactions to occur more quickly.

28. Photosynthesis is an exothermic reaction because it releases heat.

Answer: False

Explanation: Photosynthesis absorbs energy, making it endothermic.

29. Explosions are examples of extremely fast chemical reactions.

Answer: True

Explanation: They occur rapidly and release large amounts of energy in a short time.

30. A chemical reaction stops when all of the reactants have been consumed.

Answer: True

Explanation: Without reactants, no further reaction can occur.

31. Catalytic converters reduce harmful vehicle emissions by using catalysts.

Answer: True

Explanation: They convert harmful gases into less harmful substances before they are released.

32. Increasing the amount of oxygen can make combustion occur more rapidly.

Answer: True

Explanation: More oxygen increases the frequency of effective combustion reactions.

33. Galvanizing iron protects it from corrosion by coating it with zinc.

Answer: True

Explanation: Zinc acts as a protective layer and corrodes before the iron does.

Topic 7 – TRUE OR FALSE

Homeostasis

1. Homeostasis is the maintenance of a stable internal environment despite changes in the external environment.

Answer: True

Explanation: Homeostasis maintains dynamic internal balance even when external conditions change.

2. Homeostasis means the body remains completely unchanged at all times.

Answer: False

Explanation: The body constantly makes small adjustments to keep conditions within a normal range.

3. The nervous and endocrine systems are the main control systems involved in homeostasis.

Answer: True

Explanation: The nervous system responds quickly, while the endocrine system provides longer-lasting regulation.

4. Sweating when body temperature rises is an example of positive feedback.

Answer: False

Explanation: Sweating is a negative feedback mechanism because it helps return body temperature to normal.

5. The normal target value maintained by homeostasis is called the set point.

Answer: True

Explanation: Examples include normal body temperature and blood glucose level.

6. Effectors detect changes in the body's internal environment.

Answer: False

Explanation: Receptors detect changes, while effectors carry out the response.

7. Muscles and glands are examples of effectors in homeostasis.

Answer: True

Explanation: They perform actions that help restore normal conditions.

8. Negative feedback increases a change farther away from the set point.

Answer: False

Explanation: Negative feedback reverses the change and restores balance.

9. Blood glucose, blood pH, and body temperature are regulated by homeostasis.

Answer: True

Explanation: These are important internal conditions that must remain within healthy limits.

10. Human height is maintained at a constant value through homeostasis.

Answer: False

Explanation: Height naturally changes during growth and is not regulated as a homeostatic variable.

11. The hypothalamus is the primary control center for body temperature regulation.

Answer: True

Explanation: It receives information from receptors and coordinates appropriate responses.

12. When you are cold, the blood vessels in your skin widen to release more heat.

Answer: False

Explanation: They constrict to reduce heat loss.

13. Hormones are chemical messengers produced by the endocrine system.

Answer: True

Explanation: Hormones help regulate many body processes over longer periods.

14. Insulin and glucagon work together to regulate blood glucose levels.

Answer: True

Explanation: They have opposite effects that keep blood sugar within the normal range.

15. When blood sugar rises, the pancreas releases glucagon.

Answer: False

Explanation: The pancreas releases insulin to lower blood glucose levels.

16. When blood sugar falls, the pancreas releases glucagon.

Answer: True

Explanation: Glucagon stimulates the release of stored glucose from the liver.

17. The kidneys help maintain the body's water and salt balance.

Answer: True

Explanation: They adjust urine concentration to regulate fluid balance.

18. Shivering is an example of negative feedback that helps restore body temperature.

Answer: True

Explanation: Muscle contractions generate heat to raise body temperature.

19. Maintaining a constant body weight is one of the main goals of homeostasis.

Answer: False

Explanation: Body weight naturally changes and is not a fixed homeostatic set point.

20. Insulin lowers blood glucose, while glucagon raises it.

Answer: True

Explanation: These hormones have opposite actions to maintain stable blood sugar.

21. Positive feedback amplifies a change instead of reversing it.

Answer: True

Explanation: Positive feedback continues until a specific event is completed.

22. Blood clotting is an example of positive feedback.

Answer: True

Explanation: Clotting chemicals activate more clotting until the wound is sealed.

23. Homeostasis requires energy to maintain stable internal conditions.

Answer: True

Explanation: Active processes are needed to continuously regulate the body's internal environment.

24. If blood pH is not properly regulated, enzymes may stop functioning correctly.

Answer: True

Explanation: Incorrect pH can change enzyme shape and reduce their activity.

25. During exercise, breathing rate increases to help maintain oxygen and carbon dioxide balance.

Answer: True

Explanation: Faster breathing supplies more oxygen and removes excess carbon dioxide.

26. The correct order of a homeostatic feedback loop is: Stimulus → Receptor → Control Center → Effector → Response.

Answer: True

Explanation: This is the standard sequence of negative and positive feedback mechanisms.

27. Oxytocin causing stronger contractions during childbirth is an example of negative feedback.

Answer: False

Explanation: It is positive feedback because the contractions become stronger until birth occurs.

28. Long-term failure of homeostasis can lead to disease, organ damage, or death.

Answer: True

Explanation: Cells cannot function properly when internal conditions move outside safe limits.

29. Dehydration causes the kidneys to reabsorb more water into the bloodstream.

Answer: True

Explanation: This helps restore normal water balance through negative feedback.

30. When blood pressure rises above normal, the body attempts to lower it back to its set point.

Answer: True

Explanation: Negative feedback mechanisms work to return blood pressure to normal.

31. Dynamic equilibrium means the body's internal conditions never change.
Answer: False
Explanation: Dynamic equilibrium allows small fluctuations while remaining within a healthy range.

32. Diabetes is primarily associated with a failure in blood glucose regulation.
Answer: True
Explanation: It results from problems with insulin production or insulin action.

33. Without homeostasis, cells and organs cannot function properly.
Answer: True
Explanation: Failure to maintain stable internal conditions can lead to severe damage or death.

Topic 8 – TRUE OR FALSE

Evolution

1. Evolution is the gradual change in the inherited traits of a population over many generations.
Answer: True
Explanation: Evolution occurs in populations over time through the inheritance of genetic traits.

2. Evolution occurs when an individual organism changes its traits during its lifetime.
Answer: False
Explanation: Individuals may grow or adapt, but evolution occurs across generations in populations.

3. Charles Darwin proposed natural selection as the primary mechanism of evolution.
Answer: True
Explanation: Darwin introduced the theory of natural selection in *On the Origin of Species* (1859).

4. Gregor Mendel developed the theory of natural selection.
Answer: False
Explanation: Mendel is known for his work on heredity and genetics, not natural selection.

5. Natural selection favors organisms that are better adapted to their environment.
Answer: True
Explanation: Better-adapted organisms are more likely to survive and reproduce.

6. The strongest organism always survives through natural selection.
Answer: False
Explanation: Fitness refers to successful reproduction, not simply physical strength.

7. An adaptation is an inherited trait that increases an organism's chance of survival and reproduction.
Answer: True
Explanation: Adaptations improve fitness in a particular environment.

8. Variation refers to similarities among individuals of the same species.
Answer: False
Explanation: Variation describes the differences in inherited traits among individuals.

9. Heredity allows useful traits to be passed from parents to offspring.

Answer: True

Explanation: Evolution depends on inherited genetic traits.

10. Fossils are preserved remains or traces of ancient organisms.

Answer: True

Explanation: Fossils provide evidence of organisms that lived long ago.

11. Homologous structures have different evolutionary origins but identical functions.

Answer: False

Explanation: Homologous structures share a common ancestry but may have different functions.

12. The human arm, whale flipper, and bat wing are examples of homologous structures.

Answer: True

Explanation: They share the same underlying bone structure inherited from a common ancestor.

13. Vestigial structures are body parts that have lost most or all of their original function.

Answer: True

Explanation: Examples include the human appendix and tailbone.

14. Biogeography is the study of fossils found in sedimentary rocks.

Answer: False

Explanation: Biogeography studies the distribution of organisms across Earth.

15. The peppered moth is a classic example of natural selection in action.

Answer: True

Explanation: Dark-colored moths became more common during the Industrial Revolution because they were better camouflaged.

16. Artificial selection occurs when nature determines which traits are inherited.

Answer: False

Explanation: Artificial selection is controlled by humans through selective breeding.

17. Geographic isolation occurs when a physical barrier separates populations.

Answer: True

Explanation: Mountains, rivers, or oceans can prevent populations from interbreeding.

18. Bird wings and insect wings are homologous structures.

Answer: False

Explanation: They are analogous structures because they have similar functions but different evolutionary origins.

19. Analogous structures perform similar functions but evolved independently.

Answer: True

Explanation: They do not indicate a recent common ancestor.

20. Isolation can contribute to the formation of new species.

Answer: True

Explanation: Preventing gene flow allows populations to evolve independently.

21. Fossils found in deeper rock layers are generally older than fossils found in upper layers.

Answer: True

Explanation: Lower rock layers were formed earlier in Earth's history.

22. Giraffes developed long necks because they stretched them during their lifetime.

Answer: False

Explanation: Long-necked individuals survived and reproduced more successfully over many generations.

23. Choosing only the largest mangoes to plant is an example of artificial selection.

Answer: True

Explanation: Humans intentionally select desirable traits.

24. Similar early embryos of vertebrates provide evidence of common ancestry.

Answer: True

Explanation: Comparative embryology supports evolutionary relationships.

25. Reproductive isolation occurs only when a mountain or river separates populations.

Answer: False

Explanation: Reproductive isolation can also result from differences in behavior, breeding season, or mating patterns.

26. A species found naturally only in one geographic area is called endemic.

Answer: True

Explanation: The Philippine Eagle is an example of an endemic species.

27. The correct sequence for speciation is Variation → Geographic Isolation → Natural Selection → Reproductive Isolation → New Species.

Answer: True

Explanation: These processes gradually lead to the formation of new species.

28. Mutation alone consistently produces adaptations that perfectly match the environment.

Answer: False

Explanation: Mutations create variation, but natural selection determines which traits become common.

29. Vestigial pelvic bones in modern whales provide evidence that whales evolved from four-legged land mammals.

Answer: True

Explanation: Fossil evidence and comparative anatomy support this evolutionary history.

30. In science, the word "theory" means a simple guess with little supporting evidence.

Answer: False

Explanation: A scientific theory is a well-tested explanation supported by extensive evidence.